Check for updates

Scandinavian Journal of Medicine & Science in Sports

WILEY

SPECIAL ISSUE ARTICLE OPEN ACCESS

Dilemma in the Treatment of Sports Injuries in Athletes: Tendon Overuse, Muscle Strain, and Tendon Rupture

Michael Kjær[1,2] | Jesper Petersen[1] | Michael Ries Dünweber[3] | Jesper Løvind Andersen[1] | Lars Engebretsen[4,5] | Stig Peter Magnusson[1,2,6]

[1]Department of Orthopaedic Surgery, Institute of Sports Medicine Copenhagen, Copenhagen University Hospital—Bispebjerg and Frederiksberg, Copenhagen, Denmark | [2]Department of Clinical Medicine, Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark | [3]Team Danmark, Brøndby, Denmark | [4]Department of Orthopaedic Surgery, Norwegian School of Sport Sciences, University of Oslo, Oslo, Norway | [5]Oslo Sports Trauma Research Center, Oslo, Norway | [6]Department of Physical and Occupational Therapy, Copenhagen University Hospital—Bispebjerg and Frederiksberg, Copenhagen, Denmark

Correspondence: Stig Peter Magnusson (p.magnusson@sund.ku.dk)

Received: 5 January 2025 | **Revised:** 22 January 2025 | **Accepted:** 25 January 2025

Funding: The authors received no specific funding for this work.

Keywords: Achilles tendon rupture | muscle strain | tendinopathy

ABSTRACT

Injuries to the musculoskeletal system are frequent in elite sports and they are detrimental to athletic performance. This can be due to, for example, (1) overuse disorders of tendon (tendinopathy) that not only lower the training efficiency but also, in many cases, are career-ending for the athlete due to pain; (2) acute muscle strain injury that both causes prolonged absence from competition and results in many re-injuries; or (3) tendon rupture that, apart from a very long rehabilitation period, will also result in many athletes never resuming their original high level of competitive sports. For all three injury examples, several evidence-based prevention programs and treatments are available, and yet the incidence of these injuries remains high and single treatments often provide only partial recovery. In this paper, we highlight the current treatments of these three conditions and focus on the unsolved dilemmas that exist in these sports injuries.

1 | Tendon Overuse Injury (Tendinopathy)

Overuse injuries of tendons are a major challenge to performing athletes, and although the Achilles tendon and the patellar tendon are the most commonly affected sites, tendons in, for example, shoulder, foot, or elbow are also affected in sports-active individuals. Achilles tendinopathy impacts 6% of the general population and up to 50% of all endurance runners over their lifespan [1]. A meta-analysis on patellar tendinopathy in athletes has recently shown that across sports disciplines, 11% of females and 17% of males had patellar tendinopathy, and that in jumping sports like volleyball and basketball, the prevalence was 20%–35% [2–4]. Although the pathogenesis of tendinopathy is not fully elucidated, there is less documentation for it being a significant mechanical rupture and rather reflects an overload and disturbed diurnal homeostasis with accompanying swelling, hypervascularization, and gradual morphological changes of the tendon, and accompanying subjective feeling of soreness and pain.

1.1 | Current Treatment of Tendon Overuse Injury

In an overview regarding all types of treatment for Achilles tendinopathy [5], it has clearly been shown that loading exercise therapies by far surpass non-loading therapies or wait-and-see policy. Currently, muscle strength training of either concentric or eccentric nature (typical over a 12-week period) combined with a reduction in load during sports is the best evidenced treatment for tendinopathy [6–9], even when the intensity of the strength

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2025 The Author(s). *Scandinavian Journal of Medicine & Science In Sports* published by John Wiley & Sons Ltd.

training is reduced from 80%–90% 1 RM to 55% 1 RM [10]. In elite athletes, just adding controlled strength training on top of the current training bouts does not have any beneficial effect [11]. It is still an open question as to how much the reduction in regular sports exercise should be along with the recommended treatment with heavy slow resistance training, but limiting sports activity to a self-monitored pain scale of between 2 and 5 on a 10-point pain scale has been found not to limit improvement in tendinopathy symptoms over the period with strength training treatment [12]. Some data indicate that improvement in tendinopathy recovery is faster the shorter time the condition has been present. This has been shown in recreational athletes [13], whereas a similar trend was not clear in elite athletes where recovery was independent of whether the condition has been present for 1, 2, or 3 months in the athlete [14].

Platelet-rich plasma (PRP) has been used in several studies in relation to tendinopathy based on the notion that it contains numerous growth factors that are crucial for the regeneration of tendon tissue. However, well-controlled studies have shown that there is no significant effect of PRP treatment [15, 16].

Similarly, it has been shown that there is no effect of using anti-inflammatory medication in the form of oral non-steroidal anti-inflammatory drugs (NSAIDs) in both late and early stages of Achilles and patellar tendinopathy [13, 17]. This is a conundrum since there are observations of inflammatory cell accumulation in the early phase of tendinopathy [18]. However, the absence of effect may be because the drug cannot enter the vital areas of the affected tendon [19]. In lateral elbow pain, a minor positive effect of local topical gel administration of NSAID has been documented [20, 21]. Insulin-like growth factor I is known to be involved in cell proliferation, collagen production, and tissue remodeling. However, local administration of insulin-like growth factor I into the patellar tendon in combination with strength training in patients with tendinopathy did not improve the treatment outcome beyond than achieved by resistance training alone [22]. Use of local corticosteroids in the peritendinous region to cure tendinopathy has a long history and has repeatedly shown to provide a very good effect, especially in the short term [6, 23, 24]. A recent study used corticosteroids in combination with strength training in Achilles tendinopathy (while abstaining from sports participation) and could demonstrate improved clinical and physiological effects without any adverse result [25]. In contrast, in lateral elbow tendinopathy, there was no improvement with corticosteroid administration in addition to strength training [26]. High-volume injection treatment involves injecting a large volume of saline, corticosteroid, and anesthetic between the Achilles tendon and Kager's fat pad. It has been shown to have significant beneficial effects upon tendinopathy [27], but the majority of the effect was explained by the corticosteroid added to the injected volume [28]. Shock wave therapy has been shown to have a positive effect upon tendon disorder if there was an intratendinous calcification typical in the shoulder [29], and only minor or no effect was found upon non-calcified tendinopathy [30]. Laser therapy has, apart from a small pain-reducing effect, no major treatment effect on tendinopathy [31]. Finally, surgery has been tried in a number of chronic tendinopathic conditions, and on patellar tendinopathy, the effect of surgery has been shown to be equal to heavy resistance training [32] whereas a significant beneficial effect of surgery vs. conservative treatment with training and corticosteroid injection has been shown in chronic fascia plantaris tendinopathy [33].

1.2 | Unsolved Questions in the Treatment of Tendon Overuse Injury in Athletes

The signs of early development of tendinopathy need to be investigated further to prevent the exacerbation of the disorder, and likewise we need to know more regarding what signs are indicative of when to resume competitive exercise without too much risk of getting a new period of tendinopathy. When using strength training as treatment for tendinopathy, it would be helpful to know what a sufficient amount of controlled resistance exercise necessary for the treatment of tendinopathy is, and maybe even more important, how much sport-specific training can be allowed during the period of treatment with controlled heavy resistance training. Despite a good treatment effect of heavy resistance training, this does still not cure the condition 100%, and the question remains whether the combination of training with other treatments of pharmacological or physical nature improves the treatment outcome further. To what extent surgery can have any beneficial effect within the treatment of tendinopathy is yet to be documented. Finally, knowing that circadian rhythm is important for tendon homeostasis, it will be interesting to discover whether impaired homeostasis in tendinopathy can be benefited from improved circadian rhythm by applying physical exercises at a specific time point of the day.

2 | Acute Muscle Strain Injury

Explosive movements or extreme stretches of typically hamstrings or calf muscle can result in an acute rupture of the muscle-tendon complex with an abrupt discontinuation of athletic movements. Hamstrings muscle injuries are the most common muscle injury in several sports [34–36]. The most common site for these muscle strain injuries is near the myotendinous junction [37, 38]. Depending upon the severity of the rupture, the athlete is away from sports for weeks to several months [39], and often recurrent injuries at the same primary site (up to 14%–67%) can be observed when training is resumed [40, 41]. The effectiveness of rehabilitating these injuries is of major importance in efforts to avoid re-injury and for individual athletic performance and thus for success in team sports like football at the highest level [42, 43].

2.1 | Current Treatment of Acute Muscle Rupture Injury

In randomized controlled trials published on the rehabilitation of acute muscle strain injuries, the time for return to sport/play (RTS/RTP) is from around 2–12 weeks [44–51]. A review of the literature on the treatment of acute hamstring injuries documented that training rehabilitation with a focus on progressive eccentric strength exercises and gradually increasing running drills yielded the fastest return to play, and at the same time, the lowest re-injury rate [52]. It has been suggested that early onset of rehabilitation is beneficial [51], but it is debated what

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

exercises and at what time they should be initiated. Lengthening exercises have been suggested to be initiated already 5 days after injury [44, 45], but comparing initiation at 5 days compared to 16 days did not demonstrate any difference in RTS or re-injury rate [50]. This was found despite improved eccentric strength in the group that started earlier [50]. Bayer et al. [51] showed that the strength improvement during the rehabilitation period was similar in early and delayed initiation of rehabilitation, despite the early rehabilitation group resuming their full activity level 3 weeks earlier than the ones starting their rehabilitation just 1 week later.

In a study that evaluated pain during activity, maximal muscle strength, flexibility, and area of palpatory soreness in sports-active individuals with hamstring muscle strain injury, it was demonstrated that in 1/3 of the recovery period prior to RTS, the length of the area with palpatory soreness was reduced by 50% [53]. This is of clinical importance as it helps predict the overall duration of the injury recovery. Very recently, a comprehensive study of acute muscle strain injuries in the hamstrings demonstrated that two clinical tests (lower straight leg rise angle on the injured leg and discomfort during active knee extension test), two MRI characteristics (involvement of the MTJ region and low degree of anteroposterior edema), and shorter time to RTS/RTP were all independently associated with increased re-injury risk for hamstrings [54]. The discomfort during knee extension tripled the risk for re-injury, as did (×3) the demonstration of affected MTJ region with MRI. Suboptimal straight leg test only marginally increased the risk for re-injury, and edema observed on the MRI was, in fact, inversely related, that is, larger edema was associated with a smaller risk of re-injury. Further, it was found that a longer period of RTS/RTP decreased the risk for re-injury with ~1.5% per day, meaning that a prolonged rehabilitation of 1 week (7 days) potentially would decrease the risk for re-injury by around 10%, which is clinically very relevant [54]. This raises the question as to what parameters should be monitored before returning to especially elite sports with high demands. It further underlines that injuries located in the myotendinous junction are severe and require a long time before full function can be resumed.

Interestingly, it has been shown that muscle atrophy of the ruptured muscle remains up to 6 months post-injury, along with increased blood flow in the region [55]. Further, even several years after the injury, abnormal connective tissue, fat accumulation, and hypervascularization can be observed in the injured region [56]. Although strength training at a late stage after muscle strain injury can partly improve symptoms and function, it is mostly the agonist muscles that are improved and the abnormal tissue composition in previously injured muscle persist [56]. Moreover, long after injury, the injured muscle appears disorganized and has a thicker aponeurosis [57]. This indicates lasting tissue morphology changes that may be irreversible changes in the muscle–tendon unit after acute muscle strain injury.

It has been demonstrated that the use of eccentric strengthening exercises (as, e.g., Nordic hamstrings exercise) can not only reduce the risk of primary hamstring injury but will also markedly lower the risk for re-injury [58–60]. Further, it might be that a combination of different eccentric exercises even protects the hamstring muscle further against reinjuries [61], but curiously the overall number of hamstring injuries in elite football has not dramatically diminished. This may be due to low compliance and adaptation of the exercises needed in the daily training practices [59]. Other supplementary treatments after muscle strain injury have been tried, such as the administration of PRP into the region of muscle injury, which did not affect RTS or frequency of re-injury [62, 63]. Recently, the administration of protein supplementation daily for 12 weeks has been tried in individuals with calf or hamstring muscle injury, but this did also not influence RTS (Mertz unpublished observation).

2.2 | Unsolved Questions in Treatment of Acute Muscle Strain Injury in Athletes

Several clinical descriptions of the rehabilitation process have been provided, but we have little knowledge of the exact healing processes in the injured area and how we can best stimulate such a healing process to achieve an optimal outcome. Further, it is unknown to what degree the regenerating injured tissue will fully resolve after rehabilitation and when the athlete is ready to return to sports at a high level with the lowest possible risk for re-injury. Moreover, how to evaluate the "readiness" of the tissue in relation to injury rehabilitation remains unknown and if treatments other than mechanical loading (strength training, gradual onset running) can improve the healing process (e.g., inhibition of inflammation) and thus accelerate the time to become ready for sports again.

3 | Tendon Rupture Injury

Acute Achilles tendon ruptures are the most common tendon rupture in recreational and competitive athletes with a male-to-female ratio of 3:1 [64–68]. Patient-reported outcome measures (PROMs) show that sports-active individuals only reach 73–82 out of 100 points even several years after the injury, which indicates that functional recovery after tendon rupture is incomplete [20, 21]. In fact, patients are often left with permanent tendon lengthening, muscle atrophy, and weakness [69], which can result in a reduced performance level or a complete inability to return to the sport at all, that is, devastating consequences for the professional athlete [66, 70–72].

3.1 | Current Treatment of Tendon Rupture Injury

A long-standing dilemma regarding the treatment of, for example, Achilles tendon rupture has been whether to perform surgical suturing of the tendon or to treat the ruptured tendon conservatively with immobilization in a fixed plantarflexion position [73–76]. A recent large randomized controlled trial found that surgery did not result in better patient-reported outcomes compared to nonoperative treatment, but it should be noted that patients (which were recreational athletes as well as regular non-trained patients) were left with a permanent 20%–30% functional deficit irrespective of the treatment approach [76]. Importantly, both surgical and non-surgical treatments result in tendon elongation, although the non-surgical approach seems to result in an even longer tendon [77]. Importantly, the elongated tendon has been correlated to reduced physical function

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

[78, 79]. These findings speak in favor of elite athletes who want to return to high-level sports potentially benefiting more from operative rather than non-operative treatment.

Much of the existing literature on tendon length after an Achilles tendon rupture has been focused on that associated with the gastrocnemius. The free Achilles tendon is comprised of distinct tendons arising from the three triceps surae muscles [80], but prior studies of Achilles tendon rupture have only considered the length of the gastrocnemius tendon and not that of the soleus tendon. Precise measures of these distinct tendon portions are difficult, but a 3D MRI approach makes it possible to accurately assess the length [79]. Using this technique, it has been shown that there is a marked difference between the injured and uninjured soleus tendon lengths, but not the gastrocnemius tendon 1 week after surgery [81]. This suggests that the current surgical approach may not be optimal for avoiding the prolongations of the different tendons.

It has been shown that 6 weeks after surgery the tendon is elongated up to 10%–20% of the entire length of the free tendon, and in some cases as much as 50% [82]. However, it appears that the tendon continues to elongate up to 6 months after surgery, and that only 50% of the elongation occurs in the initial 3 months [83]. Importantly, the clinical outcome is inversely related to the magnitude of tendon elongation [78, 79, 84, 85]. Delayed loading after surgical repair does not impact tendon elongation or muscle function, which does not recover completely by 12 months [81]. Interestingly, the delayed rehabilitation yielded reduced tendon inflammation in the first 3 months and a greater and meaningful improvement in the **patient-reported outcome** after 1 year compared to the conventional early rehabilitation [81].

Despite extensive rehabilitation efforts, muscle weakness can persist for years and even decades, which has been attributed to a reduction in anatomical muscle cross-sectional area [77, 86, 87]. This has prompted a focus on early mobilization following the repair of the ruptured tendon to avoid muscle atrophy [88]. Yet, these efforts have proven ineffective, which is noteworthy since the triceps surae muscles can lose 30% of their mass during a 90 days bed rest and fully regain it in the following 90 days [89]. Therefore, immobilization-associated anatomical cross-sectional muscle atrophy, per se, is an unlikely mechanism for the permanent loss of muscle mass and excursion after Achilles tendon rupture.

Impaired range of motion during a heel rise has been correlated to tendon elongation after an Achilles tendon rupture, which appears to be associated with a shorter muscle [79]. The number of parallel sarcomeres governs the force-generating capacity, while the number of sarcomeres in series governs the muscle's total excursion, i.e., joint range of motion [90]. If the tendon is elongated and the muscle is consequently shortened, the force-generating capacity could theoretically be maintained if the serial sarcomere number adjusts to maintain the optimal actin-myosin overlap. However, the shorter muscle would generate less force closer to the maximal plantarflexion angle [91], which is typically seen clinically [81, 91, 92], Recently, it was shown that Achilles tendon elongation results in a loss of muscle mass and length with an accompanying normalization in serial sarcomere count in an animal model [93]. It is possible that the tendon elongation that is accompanied by a shortening of the muscle can explain the reduced end-range muscle strength and heel-rise height.

3.2 | Unsolved Questions in the Treatment of Tendon Rupture in Athletes

Although there is some evidence that a surgical approach is preferential for elite athletes, the optimal treatment remains an enigma, and several questions are unanswered. For example, is there a surgical technique that can restore the anatomical configuration and length of the Achilles tendons' sub-sections? Further, when is the best time to initiate loading to avoid tendon elongation and muscle shortening, and how should this loading be progressed? Associated with this rehabilitation program, how can we best monitor tendon elongation and tissue inflammation during rehabilitation noninvasively.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

This is a narrative review.

References

1. U. M. Kujala, S. Sarna, and J. Kaprio, "Cumulative Incidence of Achilles Tendon Rupture and Tendinopathy in Male Former Elite Athletes," *Clinical Journal of Sport Medicine* 15, no. 3 (2005): 133–135.

2. S. Nutarelli, C. M. T. da Lodi, J. L. Cook, L. Deabate, and G. Filardo, "Epidemiology of Patellar Tendinopathy in Athletes and the General Population: A Systematic Review and Meta-Analysis," *Orthopedics & Sports Medicine* 11, no. 6 (2023): 23259671231173659.

3. M. Hagglund, J. Zwerver, and J. Ekstrand, "Epidemiology of Patellar Tendinopathy in Elite Male Soccer Players," *American Journal of Sports Medicine* 39 (2011): 1906–1911.

4. M. K. Hutchison, J. Houck, T. Cuddeford, R. Dorociak, and J. Brumitt, "Prevalence of Patellar Tendinopathy and Patellar Tendon Abnormality in Male Collegiate Basketball Players: A Cross-Sectional Study," *Journal of Athletic Training* 54, no. 9 (2019): 953–958.

5. A. C. van der Vlist, M. Winters, A. Weir, et al., "Which Treatment Is Most Effective for Patients With Achilles Tendinopathy? A Living Systematic Review With Network Meta-Analysis of 29 Randomised Controlled Trials," *British Journal of Sports Medicine* 55, no. 5 (2021): 249–256.

6. M. Kongsgaard, V. Kovanen, P. Aagaard, et al., "Corticosteroid Injections, Eccentric Decline Squat Training and Heavy Slow Resistance Training in Patellar Tendinopathy," *Scandinavian Journal of Medicine & Science in Sports* 19, no. 6 (2009): 790–802.

7. R. Beyer, M. Kongsgaard, B. Hougs Kjaer, T. Ohlenschlaeger, M. Kjaer, and S. P. Magnusson, "Heavy Slow Resistance Versus Eccentric Training as Treatment for Achilles Tendinopathy: A Randomized Controlled Trial," *American Journal of Sports Medicine* 43, no. 7 (2015): 1704–1711.

8. A. Frohm, T. Saartok, K. Halvorsen, and P. Renstrom, "Eccentric Treatment for Patellar Tendinopathy: A Prospective Randomised Short-Term Pilot Study of Two Rehabilitation Protocols," *British Journal of Sports Medicine* 41, no. 7 (2007): e7.

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

9. P. Jonsson and H. Alfredson, "Superior Results With Eccentric Compared to Concentric Quadriceps Training in Patients With Jumper's Knee: A Prospective Randomised Study," *British Journal of Sports Medicine* 39, no. 11 (2005): 847–850.

10. A. S. Agergaard, R. B. Svensson, N. M. Malmgaard-Clausen, et al., "Clinical Outcomes, Structure, and Function Improve With Both Heavy and Moderate Loads in the Treatment of Patellar Tendinopathy: A Randomized Clinical Trial," *American Journal of Sports Medicine* 49, no. 4 (2021): 982–993.

11. H. Visnes, A. Hoksrud, J. Cook, and R. Bahr, "No Effect of Eccentric Training on Jumper's Knee in Volleyball Players During the Competitive Season: A Randomized Clinical Trial," *Clinical Journal of Sport Medicine* 15, no. 4 (2005): 227–234.

12. K. G. Silbernagel, R. Thomee, B. I. Eriksson, and J. Karlsson, "Continued Sports Activity, Using a Pain-Monitoring Model, During Rehabilitation in Patients With Achilles Tendinopathy: A Randomized Controlled Study," *American Journal of Sports Medicine* 35, no. 6 (2007): 897–906.

13. N. M. Malmgaard-Clausen, O. H. Jorgensen, R. Hoffner, et al., "No Additive Clinical or Physiological Effects of Short-Term Anti-Inflammatory Treatment to Physical Rehabilitation in the Early Phase of Human Achilles Tendinopathy: A Randomized Controlled Trial," *American Journal of Sports Medicine* 49, no. 7 (2021): 1711–1720.

14. C. S. Meulengracht, M. Seidler, R. B. Svensson, et al., "Clinical and Imaging Outcomes Over 12 Weeks in Elite Athletes With Early-Stage Tendinopathy," *Scandinavian Journal of Medicine & Science in Sports* 34, no. 10 (2024): e14732.

15. R. J. de Vos, A. Weir, H. T. van Schie, et al., "Platelet-Rich Plasma Injection for Chronic Achilles Tendinopathy: A Randomized Controlled Trial," *Journal of the American Medical Association* 303, no. 2 (2010): 144–149.

16. R. S. Kearney, C. Ji, J. Warwick, et al., "Effect of Platelet-Rich Plasma Injection vs Sham Injection on Tendon Dysfunction in Patients With Chronic Midportion Achilles Tendinopathy: A Randomized Clinical Trial," *Journal of the American Medical Association* 326, no. 2 (2021): 137–144.

17. M. Astrom and N. Westlin, "No Effect of Piroxicam on Achilles Tendinopathy. A Randomized Study of 70 Patients," *Acta Orthopaedica Scandinavica* 63, no. 6 (1992): 631–634.

18. B. J. Dean, P. Gettings, S. G. Dakin, and A. J. Carr, "Are Inflammatory Cells Increased in Painful Human Tendinopathy? A Systematic Review," *British Journal of Sports Medicine* 50, no. 4 (2015): 216–220, https://doi.org/10.1136/bjsports-2015-094754.

19. K. M. Heinemeier, T. F. Ohlenschlaeger, U. R. Mikkelsen, et al., "Effects of Anti-Inflammatory (NSAID) Treatment on Human Tendinopathic Tissue," *Journal of Applied Physiology* 123, no. 5 (2017): 1397–1405.

20. P. Pattanittum, T. Turner, S. Green, and R. Buchbinder, "Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) for Treating Lateral Elbow Pain in Adults," *Cochrane Database of Systematic Reviews* 2013, no. 5 (2013): CD003686.

21. L. A. N. Crowe, M. Akbar, R. J. de Vos, et al., "Pathways Driving Tendinopathy and Enthesitis: Siblings or Distant Cousins in Musculoskeletal Medicine?," *Lancet Rheumatology* 5, no. 5 (2023): e293–e304.

22. J. L. Olesen, M. Hansen, I. F. Turtumoygard, et al., "No Treatment Benefits of Local Administration of Insulin-Like Growth Factor-1 in Addition to Heavy Slow Resistance Training in Tendinopathic Human Patellar Tendons: A Randomized, Double-Blind, Placebo-Controlled Trial With 1-Year Follow-Up," *American Journal of Sports Medicine* 49, no. 9 (2021): 3635465211021056, https://doi.org/10.1177/03635465211021056.

23. T. P. Krogh, U. Fredberg, K. Stengaard-Pedersen, R. Christensen, P. Jensen, and T. Ellingsen, "Treatment of Lateral Epicondylitis With Platelet-Rich Plasma, Glucocorticoid, or Saline: A Randomized, Double-Blind, Placebo-Controlled Trial," *American Journal of Sports Medicine* 41, no. 3 (2013): 625–635.

24. N. Smidt and D. A. van der Windt, "Tennis Elbow in Primary Care," *British Medical Journal* 333, no. 7575 (2006): 927–928.

25. F. Johannsen, J. L. Olesen, T. F. Ohlenschlager, et al., "Effect of Ultrasonography-Guided Corticosteroid Injection vs Placebo Added to Exercise Therapy for Achilles Tendinopathy: A Randomized Clinical Trial," *JAMA Network Open* 5, no. 7 (2022): e2219661.

26. C. Couppe, S. Dossing, P. M. Bulow, et al., "Effects of Heavy Slow Resistance Training Combined With Corticosteroid Injections or Tendon Needling in Patients With Lateral Elbow Tendinopathy: A 3-Arm Randomized Double-Blinded Placebo-Controlled Study," *American Journal of Sports Medicine* 50 (2022): 3635465221110214.

27. A. P. Boesen, R. Hansen, M. I. Boesen, P. Malliaras, and H. Langberg, "Effect of High-Volume Injection, Platelet-Rich Plasma, and Sham Treatment in Chronic Midportion Achilles Tendinopathy: A Randomized Double-Blinded Prospective Study," *American Journal of Sports Medicine* 45, no. 9 (2017): 2034–2043.

28. A. P. Boesen, H. Langberg, R. Hansen, P. Malliaras, and M. I. Boesen, "High Volume Injection With and Without Corticosteroid in Chronic Midportion Achilles Tendinopathy," *Scandinavian Journal of Medicine & Science in Sports* 29, no. 8 (2019): 1223–1231.

29. L. Gerdesmeyer, S. Wagenpfeil, M. Haake, et al., "Extracorporeal Shock Wave Therapy for the Treatment of Chronic Calcifying Tendonitis of the Rotator Cuff: A Randomized Controlled Trial," *Journal of the American Medical Association* 290, no. 19 (2003): 2573–2580.

30. V. Korakakis, R. Whiteley, A. Tzavara, and N. Malliaropoulos, "The Effectiveness of Extracorporeal Shockwave Therapy in Common Lower Limb Conditions: A Systematic Review Including Quantification of Patient-Rated Pain Reduction," *British Journal of Sports Medicine* 52, no. 6 (2018): 387–407.

31. I. F. Naterstad, J. Joensen, J. M. Bjordal, C. Couppe, R. A. B. Lopes-Martins, and M. B. Stausholm, "Efficacy of Low-Level Laser Therapy in Patients With Lower Extremity Tendinopathy or Plantar Fasciitis: Systematic Review and Meta-Analysis of Randomised Controlled Trials," *BMJ Open* 12, no. 9 (2022): e059479.

32. R. Bahr, B. Fossan, S. Loken, and L. Engebretsen, "Surgical Treatment Compared With Eccentric Training for Patellar Tendinopathy (Jumper's Knee). A Randomized, Controlled Trial," *Journal of Bone and Joint Surgery. American Volume* 88, no. 8 (2006): 1689–1698.

33. F. Johannsen, L. Konradsen, R. Herzog, and M. R. Krogsgaard, "Endoscopic Fasciotomy for Plantar Fasciitis Provides Superior Results When Compared to a Controlled Non-Operative Treatment Protocol: A Randomized Controlled Trial," *Knee Surgery, Sports Traumatology, Arthroscopy* 28, no. 10 (2020): 3301–3308.

34. P. Edouard, P. Branco, and J. M. Alonso, "Muscle Injury Is the Principal Injury Type and Hamstring Muscle Injury Is the First Injury Diagnosis During Top-Level International Athletics Championships Between 2007 and 2015," *British Journal of Sports Medicine* 50, no. 10 (2016): 619–630.

35. J. Ekstrand, H. Bengtsson, M. Walden, M. Davison, K. M. Khan, and M. Hagglund, "Hamstring Injury Rates Have Increased During Recent Seasons and Now Constitute 24% of All Injuries in Men's Professional Football: The UEFA Elite Club Injury Study From 2001/02 to 2021/22," *British Journal of Sports Medicine* 57, no. 5 (2022): 292–298.

36. J. Petersen, K. Thorborg, M. B. Nielsen, and P. Holmich, "Acute Hamstring Injuries in Danish Elite Football: A 12-Month Prospective Registration Study Among 374 Players," *Scandinavian Journal of Medicine & Science in Sports* 20, no. 4 (2010): 588–592.

37. J. G. Tidball, G. Salem, and R. Zernicke, "Site and Mechanical Conditions for Failure of Skeletal Muscle in Experimental Strain Injuries," *Journal of Applied Physiology* 74, no. 3 (1993): 1280–1286.

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

38. A. B. Knudsen, M. Larsen, A. L. Mackey, et al., "The Human Myotendinous Junction: An Ultrastructural and 3D Analysis Study," *Scandinavian Journal of Medicine & Science in Sports* 25, no. 1 (2015): e116–e123.

39. J. Hickey, A. J. Shield, M. D. Williams, and D. A. Opar, "The Financial Cost of Hamstring Strain Injuries in the Australian Football League," *British Journal of Sports Medicine* 48, no. 8 (2014): 729–730.

40. H. M. de Visser, M. Reijman, M. P. Heijboer, and P. K. Bos, "Risk Factors of Recurrent Hamstring Injuries: A Systematic Review," *British Journal of Sports Medicine* 46, no. 2 (2012): 124–130.

41. A. Wangensteen, J. L. Tol, E. Witvrouw, et al., "Hamstring Reinjuries Occur at the Same Location and Early After Return to Sport: A Descriptive Study of MRI-Confirmed Reinjuries," *American Journal of Sports Medicine* 44, no. 8 (2016): 2112–2121.

42. M. Hagglund, M. Walden, and J. Ekstrand, "Injury Recurrence Is Lower at the Highest Professional Football Level Than at National and Amateur Levels: Does Sports Medicine and Sports Physiotherapy Deliver?," *British Journal of Sports Medicine* 50, no. 12 (2016): 751–758.

43. H. Bengtsson, J. Ekstrand, M. Walden, and M. Hagglund, "Match Injury Rates in Professional Soccer Vary With Match Result, Match Venue, and Type of Competition," *American Journal of Sports Medicine* 41, no. 7 (2013): 1505–1510.

44. C. M. Askling, M. Tengvar, and A. Thorstensson, "Acute Hamstring Injuries in Swedish Elite Football: A Prospective Randomised Controlled Clinical Trial Comparing Two Rehabilitation Protocols," *British Journal of Sports Medicine* 47, no. 15 (2013): 953–959.

45. C. M. Askling, M. Tengvar, O. Tarassova, and A. Thorstensson, "Acute Hamstring Injuries in Swedish Elite Sprinters and Jumpers: A Prospective Randomised Controlled Clinical Trial Comparing Two Rehabilitation Protocols," *British Journal of Sports Medicine* 48, no. 7 (2014): 532–539.

46. J. Mendiguchia, E. Martinez-Ruiz, P. Edouard, et al., "A Multifactorial, Criteria-Based Progressive Algorithm for Hamstring Injury Treatment," *Medicine and Science in Sports and Exercise* 49, no. 7 (2017): 1482–1492.

47. M. A. Sherry and T. M. Best, "A Comparison of 2 Rehabilitation Programs in the Treatment of Acute Hamstring Strains," *Journal of Orthopaedic & Sports Physical Therapy* 34, no. 3 (2004): 116–125.

48. A. Silder, M. A. Sherry, J. Sanfilippo, M. J. Tuite, S. J. Hetzel, and B. C. Heiderscheit, "Clinical and Morphological Changes Following 2 Rehabilitation Programs for Acute Hamstring Strain Injuries: A Randomized Clinical Trial," *Journal of Orthopaedic & Sports Physical Therapy* 43, no. 5 (2013): 284–299.

49. J. T. Hickey, R. G. Timmins, N. Maniar, et al., "Pain-Free Versus Pain-Threshold Rehabilitation Following Acute Hamstring Strain Injury: A Randomized Controlled Trial," *Journal of Orthopaedic and Sports Physical Therapy* 50, no. 2 (2020): 91–103.

50. R. Vermeulen, R. Whiteley, A. D. van der Made, et al., "Early Versus Delayed Lengthening Exercises for Acute Hamstring Injury in Male Athletes: A Randomised Controlled Clinical Trial," *British Journal of Sports Medicine* 56, no. 14 (2022): 792–800.

51. M. L. Bayer, S. P. Magnusson, M. Kjaer, and Tendon Research Group B, "Early Versus Delayed Rehabilitation After Acute Muscle Injury," *New England Journal of Medicine* 377, no. 13 (2017): 1300–1301.

52. L. Ishoi, K. Krommes, R. S. Husted, C. B. Juhl, and K. Thorborg, "Diagnosis, Prevention and Treatment of Common Lower Extremity Muscle Injuries in Sport—Grading the Evidence: A Statement Paper Commissioned by the Danish Society of Sports Physical Therapy (DSSF)," *British Journal of Sports Medicine* 54, no. 9 (2020): 528–537.

53. R. Whiteley, N. van Dyk, A. Wangensteen, and C. Hansen, "Clinical Implications From Daily Physiotherapy Examination of 131 Acute Hamstring Injuries and Their Association With Running Speed and Rehabilitation Progression," *British Journal of Sports Medicine* 52, no. 5 (2018): 303–310.

54. M. I. Zein, M. J. K. Mokkenstorm, M. Cardinale, et al., "Baseline Clinical and MRI Risk Factors for Hamstring Reinjury Showing the Value of Performing Baseline MRI and Delaying Return to Play: A Multicentre, Prospective Cohort of 330 Acute Hamstring Injuries," *British Journal of Sports Medicine* 58, no. 14 (2024): 766–776.

55. M. L. Bayer, M. Hoegberget-Kalisz, M. H. Jensen, et al., "Role of Tissue Perfusion, Muscle Strength Recovery, and Pain in Rehabilitation After Acute Muscle Strain Injury: A Randomized Controlled Trial Comparing Early and Delayed Rehabilitation," *Scandinavian Journal of Medicine & Science in Sports* 28 (2018): 2579–2591.

56. M. L. Bayer, M. Hoegberget-Kalisz, R. B. Svensson, et al., "Chronic Sequelae After Muscle Strain Injuries: Influence of Heavy Resistance Training on Functional and Structural Characteristics in a Randomized Controlled Trial," *American Journal of Sports Medicine* 49, no. 10 (2021): 3635465211026623, https://doi.org/10.1177/03635465211026623.

57. L. B Nielsen, R. B Svensson, N. U Fredskild, et al., "Chronic Changes in Muscle Architecture and Aponeurosis Structure Following Calf Muscle Strain Injuries," *Scandinavian Journal of Medicine & Science in Sports* 33, no. 12 (2023): 2585–2597.

58. J. Petersen, K. Thorborg, M. B. Nielsen, E. Budtz-Jorgensen, and P. Holmich, "Preventive Effect of Eccentric Training on Acute Hamstring Injuries in Men's Soccer: A Cluster-Randomized Controlled Trial," *American Journal of Sports Medicine* 39, no. 11 (2011): 2296–2303.

59. J. Ekstrand, H. Bengtsson, M. Walden, M. Davison, and M. Hagglund, "Still Poorly Adopted in Male Professional Football: But Teams That Used the Nordic Hamstring Exercise in Team Training Had Fewer Hamstring Injuries—A Retrospective Survey of 17 Teams of the UEFA Elite Club Injury Study During the 2020–2021 Season," *BMJ Open Sport & Exercise Medicine* 8, no. 3 (2022): e001368.

60. A. Arnason, T. E. Andersen, I. Holme, L. Engebretsen, and R. Bahr, "Prevention of Hamstring Strains in Elite Soccer: An Intervention Study," *Scandinavian Journal of Medicine & Science in Sports* 18, no. 1 (2008): 40–48.

61. J. J. M. Suskens, L. Secondulfo, O. Kilic, et al., "Effect of Two Eccentric Hamstring Exercises on Muscle Architectural Characteristics Assessed With Diffusion Tensor MRI," *Scandinavian Journal of Medicine & Science in Sports* 33, no. 4 (2023): 393–406.

62. G. Reurink, G. J. Goudswaard, M. H. Moen, et al., "Platelet-Rich Plasma Injections in Acute Muscle Injury," *New England Journal of Medicine* 370, no. 26 (2014): 2546–2547.

63. B. Hamilton, J. L. Tol, E. Almusa, et al., "Platelet-Rich Plasma Does Not Enhance Return to Play in Hamstring Injuries: A Randomised Controlled Trial," *British Journal of Sports Medicine* 49, no. 14 (2015): 943–950.

64. S. A. Nillius, B. E. Nilsson, and N. E. Westlin, "The Incidence of Achilles Tendon Rupture," *Acta Orthopaedica Scandinavica* 47, no. 1 (1976): 118–121.

65. A. Ganestam, T. Kallemose, A. Troelsen, and K. W. Barfod, "Increasing Incidence of Acute Achilles Tendon Rupture and a Noticeable Decline in Surgical Treatment From 1994 to 2013. A Nationwide Registry Study of 33,160 Patients," *Knee Surgery, Sports Traumatology, Arthroscopy* 24, no. 12 (2016): 3730–3737.

66. A. Grassi, S. Caravelli, M. Fuiano, et al., "Epidemiology of Achilles Tendon Rupture in Italian First Division Football (Soccer) Players and Their Performance After Return to Play," *Clinical Journal of Sport Medicine* 32, no. 1 (2022): e90–e95.

67. I. Lantto, J. Heikkinen, T. Flinkkila, P. Ohtonen, and J. Leppilahti, "Epidemiology of Achilles Tendon Ruptures: Increasing Incidence Over a 33-Year Period," *Scandinavian Journal of Medicine & Science in Sports* 25, no. 1 (2015): e133–e138.

68. N. J. Lemme, N. Y. Li, S. F. DeFroda, J. Kleiner, and B. D. Owens, "Epidemiology of Achilles Tendon Ruptures in the United States: Athletic and Nonathletic Injuries From 2012 to 2016," *Orthopaedic Journal of Sports Medicine* 6, no. 11 (2018): 2325967118808238.

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

69. R. Hoeffner, R. B. Svensson, N. Bjerregaard, M. Kjaer, and S. P. Magnusson, "Persistent Deficits After an Achilles Tendon Rupture: A Narrative Review," *Translational Sports Medicine* 2022 (2022): 7445398, https://doi.org/10.1155/2022/7445398.

70. D. P. Trofa, J. C. Miller, E. S. Jang, D. R. Woode, J. K. Greisberg, and J. T. Vosseller, "Professional Athletes' Return to Play and Performance After Operative Repair of an Achilles Tendon Rupture," *American Journal of Sports Medicine* 45 (2017): 363546517713001.

71. N. H. Amin, A. B. Old, L. P. Tabb, R. Garg, N. Toossi, and D. L. Cerynik, "Performance Outcomes After Repair of Complete Achilles Tendon Ruptures in National Basketball Association Players," *American Journal of Sports Medicine* 41 (2013): 1864–1868.

72. W. Johns, K. C. Walley, R. Seedat, D. B. Thordarson, B. Jackson, and T. Gonzalez, "Career Outlook and Performance of Professional Athletes After Achilles Tendon Rupture: A Systematic Review," *Foot and Ankle International* 42, no. 4 (2021): 495–509.

73. K. Nilsson-Helander, K. G. Silbernagel, R. Thomee, et al., "Acute Achilles Tendon Rupture: A Randomized, Controlled Study Comparing Surgical and Nonsurgical Treatments Using Validated Outcome Measures," *American Journal of Sports Medicine* 38, no. 11 (2010): 2186–2193.

74. N. Olsson, K. G. Silbernagel, B. I. Eriksson, et al., "Stable Surgical Repair With Accelerated Rehabilitation Versus Nonsurgical Treatment for Acute Achilles Tendon Ruptures: A Randomized Controlled Study," *American Journal of Sports Medicine* 41, no. 12 (2013): 2867–2876.

75. Y. Ochen, R. B. Beks, M. van Heijl, et al., "Operative Treatment Versus Nonoperative Treatment of Achilles Tendon Ruptures: Systematic Review and Meta-Analysis," *British Medical Journal* 364 (2019): k5120.

76. S. B. Myhrvold, E. F. Brouwer, T. K. M. Andresen, et al., "Nonoperative or Surgical Treatment of Acute Achilles' Tendon Rupture," *New England Journal of Medicine* 386, no. 15 (2022): 1409–1420.

77. J. Heikkinen, I. Lantto, T. Flinkkila, et al., "Soleus Atrophy Is Common After the Nonsurgical Treatment of Acute Achilles Tendon Ruptures: A Randomized Clinical Trial Comparing Surgical and Nonsurgical Functional Treatments," *American Journal of Sports Medicine* 45, no. 6 (2017): 1395–1404.

78. K. G. Silbernagel, R. Steele, and K. Manal, "Deficits in Heel-Rise Height and Achilles Tendon Elongation Occur in Patients Recovering From an Achilles Tendon Rupture," *American Journal of Sports Medicine* 40, no. 7 (2012): 1564–1571.

79. R. B. Svensson, C. Couppé, A. S. Agergaard, et al., "Persistent Functional Loss Following Ruptured Achilles Tendon Is Associated With Reduced Gastrocnemius Muscle Fascicle Length, Elongated Gastrocnemius and Soleus Tendon, and Reduced Muscle Cross-Sectional Area," *Translational Sports Medicine* 2, no. 6 (2019): 316–324.

80. J. Bojsen-Moller and S. P. Magnusson, "Mechanical Properties, Physiological Behavior and Function of Aponeurosis and Tendon," *Journal of Applied Physiology* 126 (2019): 1800–1807.

81. R. Hoeffner, A. S. Agergaard, R. B. Svensson, et al., "Tendon Elongation and Function After Delayed or Standard Loading of Surgically Repaired Achilles Tendon Ruptures: A Randomized Controlled Trial," *American Journal of Sports Medicine* 52, no. 4 (2024): 1022–1031.

82. H. M. Mortensen, O. Skov, and P. E. Jensen, "Early Motion of the Ankle After Operative Treatment of a Rupture of the Achilles Tendon. A Prospective, Randomized Clinical and Radiographic Study," *Journal of Bone and Joint Surgery. American Volume* 81, no. 7 (1999): 983–990.

83. P. Eliasson, A. S. Agergaard, C. Couppe, et al., "The Ruptured Achilles Tendon Elongates for 6 Months After Surgical Repair Regardless of Early or Late Weightbearing in Combination With Ankle Mobilization: A Randomized Clinical Trial," *American Journal of Sports Medicine* 46, no. 10 (2018): 2492–2502.

84. J. Kangas, A. Pajala, P. Ohtonen, and J. Leppilahti, "Achilles Tendon Elongation After Rupture Repair: A Randomized Comparison of 2 Postoperative Regimens," *American Journal of Sports Medicine* 35, no. 1 (2007): 59–64.

85. J. Heikkinen, I. Lantto, J. Piilonen, et al., "Tendon Length, Calf Muscle Atrophy, and Strength Deficit After Acute Achilles Tendon Rupture: Long-Term Follow-Up of Patients in a Previous Study," *Journal of Bone and Joint Surgery. American Volume* 99, no. 18 (2017): 1509–1515.

86. T. Haggmark, H. Liedberg, E. Eriksson, and T. Wredmark, "Calf Muscle Atrophy and Muscle Function After Non-Operative vs Operative Treatment of Achilles Tendon Ruptures," *Orthopedics* 9, no. 2 (1986): 160–164.

87. J. Leppilahti, S. Lahde, K. Forsman, J. Kangas, K. Kauranen, and S. Orava, "Relationship Between Calf Muscle Size and Strength After Achilles Rupture Repair," *Foot and Ankle International* 21, no. 4 (2000): 330–335.

88. M. Brumann, S. F. Baumbach, W. Mutschler, and H. Polzer, "Accelerated Rehabilitation Following Achilles Tendon Repair After Acute Rupture—Development of an Evidence-Based Treatment Protocol," *Injury* 45, no. 11 (2014): 1782–1790.

89. D. L. Belavy, H. Ohshima, J. Rittweger, and D. Felsenberg, "High-Intensity Flywheel Exercise and Recovery of Atrophy After 90 Days Bed–Rest," *BMJ Open Sport & Exercise Medicine* 3, no. 1 (2017): e000196.

90. A. F. Huxley and R. Niedergerke, "Structural Changes in Muscle During Contraction; Interference Microscopy of Living Muscle Fibres," *Nature* 173, no. 4412 (1954): 971–973.

91. B. Staudle, O. Seynnes, G. Laps, F. Goll, G. P. Bruggemann, and K. Albracht, "Recovery From Achilles Tendon Repair: A Combination of Postsurgery Outcomes and Insufficient Remodeling of Muscle and Tendon," *Medicine and Science in Sports and Exercise* 53, no. 7 (2021): 1356–1366.

92. M. J. Mullaney, M. P. McHugh, T. F. Tyler, S. J. Nicholas, and S. J. Lee, "Weakness in End-Range Plantar Flexion After Achilles Tendon Repair," *American Journal of Sports Medicine* 34, no. 7 (2006): 1120–1125.

93. R. Hoeffner, R. B. Svensson, F. Dietrich-Zagonel, et al., "Muscle Fascicle and Sarcomere Adaptation in Response to Achilles Tendon Elongation in an Animal Model," *Journal of Applied Physiology* 135, no. 2 (2023): 326–333.

16000838, 2025, 2, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/sms.70026, Wiley Online Library on [21/05/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License